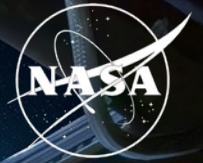


National Aeronautics and  
Space Administration



# Interpretable and generative models for mouse spaceflight

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Biological & Physical Sciences



## PROJECT

# One dataset, two machine-learning questions

We used mouse bulk RNA-seq to study how spaceflight changes genes, pathways, and tissue state.

**NASA OSDR API** Mus musculus bulk RNA-seq | spaceflight (FLT) and ground control (GC) | multiple missions and tissues

## 01 Which biological programs change?

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### expiMap

Train a pathway-constrained reference on ARCHS4, then map OSDR samples into mouse Reactome programs.

**Output: tissue-specific pathway shifts**

## 02 Can synthetic profiles sharpen the analysis?

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### Conditional generation

Generate matched expression profiles by tissue, study context, and FLT or GC condition.

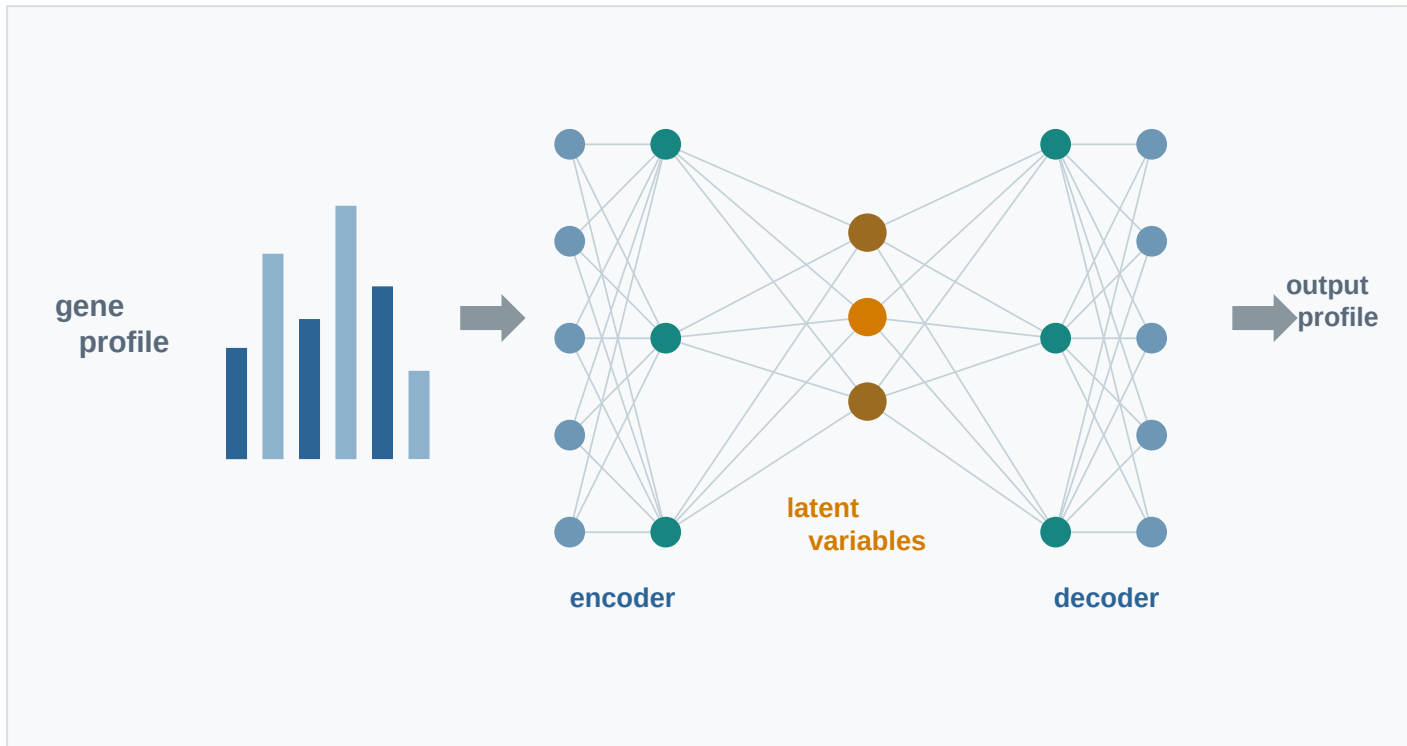
**Output: validated profiles and gene ranking**



# Autoencoders compress expression into a latent space

The encoder summarizes a gene profile; the decoder learns enough structure to reconstruct it.

## AUTOENCODER



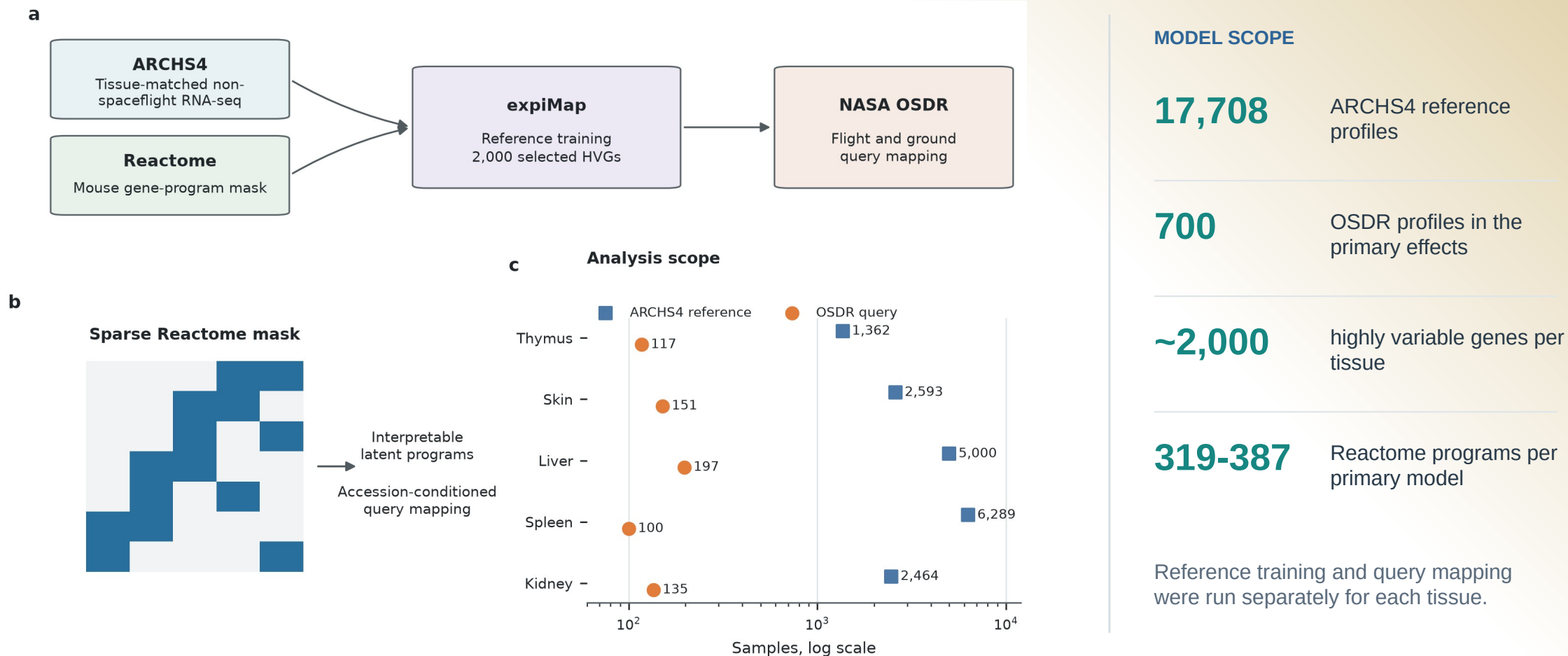
## LATENT SPACE



**expiMap:** each constrained latent variable corresponds to a Reactome gene program.

# Reactome pathways make the latent space interpretable

Tissue-matched ARCHS4 references define the background; OSDR FLT and GC samples are mapped as queries.



## EXPIMAP RESULTS

# Four tissues produced the clearest pathway patterns

Effects were estimated within each project before the project-level shifts were combined.

## Thymus

117 samples | 5 projects

### LOWER IN FLIGHT

**DNA repair; RHOA cytoskeletal cycle; lymphoid-stromal interactions**

Known involution may also involve lower repair, motility, and niche coordination.

## Skin

151 samples | 4 projects

### LOWER IN FLIGHT

**Chromatin regulation; DNA repair; Hedgehog; sphingolipids; cell junctions**

Barrier injury may include a broader loss of tissue maintenance and coordination.

## Liver

197 samples | 9 projects

### LOWER IN FLIGHT

**MHC class II antigen presentation; T-cell receptor signaling**

Metabolic heterogeneity coexisted with lower adaptive immune communication.

## Spleen

100 samples | 5 projects

### LOWER IN FLIGHT

**T-cell receptor signaling; neutrophil degranulation; C-type lectin signaling**

The most consistent multi-pathway result joined adaptive and innate immune changes.

## BACKGROUND

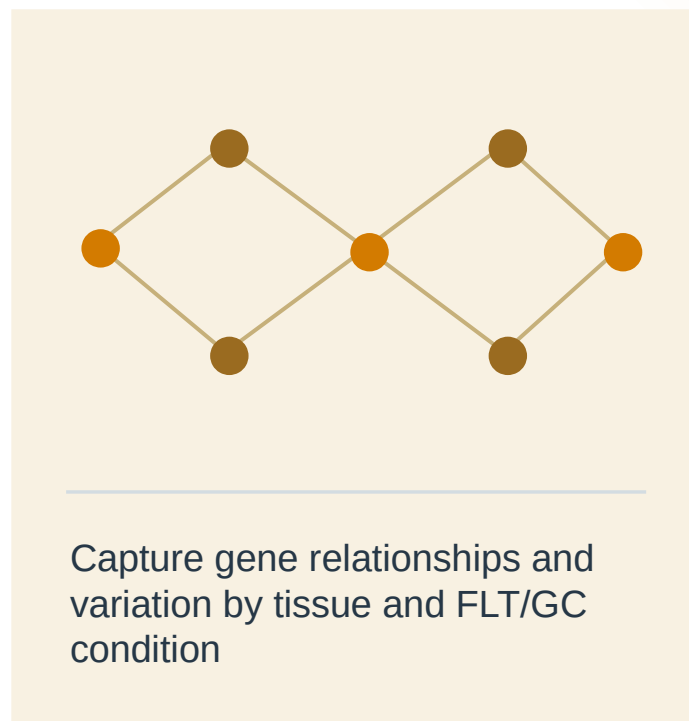
# What is synthetic transcriptomics?

A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

## 01 Observed profiles



## 02 Learn the distribution



## 03 Synthetic profiles



## QUESTION

# Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

## NASA OSDR

Observed spaceflight cohort

**1,610**

profiles

**75**

accessions

**835 FLT**

**775 GC**



## ARCHS4 mouse

Reference pretraining cohort

**997,515**

profiles screened



**17,244**

selected

**20 tissues**

## Study effects can look like spaceflight biology.

- 01 Preserve tissue and FLT/GC structure.
- 02 Avoid memorizing individual profiles.
- 03 Test biological effects in observed OSDR samples.

## METHOD

# We built a configurable bulk RNA-seq generation pipeline

The downstream analysis used the outlined options; the remaining choices stay configurable.

## 01 Data scope

### Sources

OSDR API

ARCHS4

### Studies

Single

Multiple

### Tissues

Per tissue

All tissues

## 02 Transform

### Expression

Raw

CPM

TPM

### Scaling

None

Z-score

MaxAbs

### Features

All

HVG

L1000

## 03 Harmonization

### Global or study correction

None

Within-study z-score

ComBat / MBatch

MOBER

## 04 Model training

### Generator

WGAN-GP

DDIM

### Training source

OSDR only

ARCHS4 only

Pretrain + adapt

## 05 Conditioning

### Model inputs

Tissue

FLT / GC

Accession

Material type

Sex / age

## Downstream

configuration

### PREPROCESS

TPM / MaxAbs / 974 landmarks

### PRETRAIN

ARCHS4

### ADAPT

OSDR

### GENERATE

Conditional DDIM

### CONDITION

Tissue

FLT / GC

Accession

Material type

### HARMONIZE

None

### SCOPE

All tissues

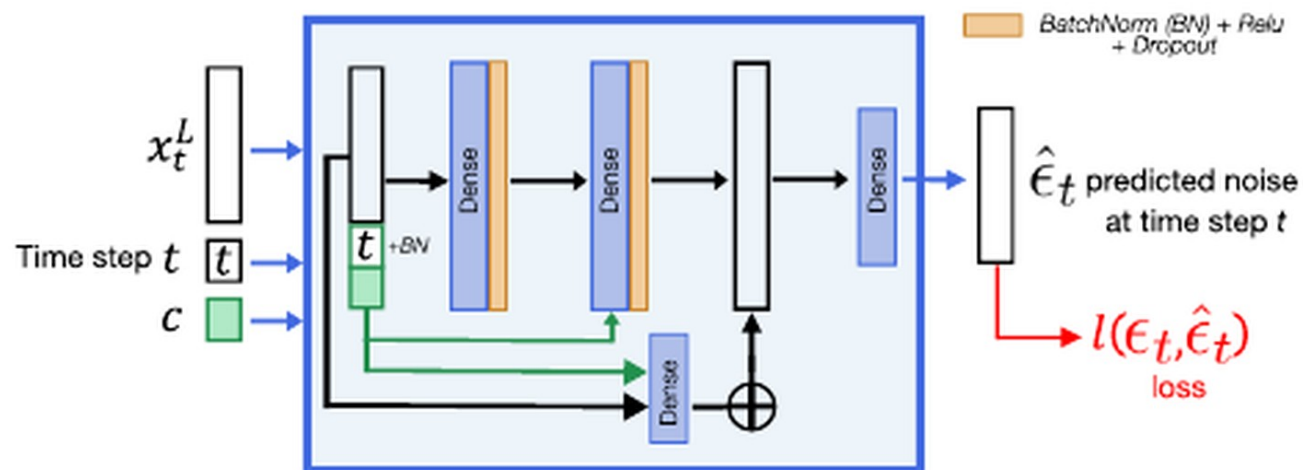


## VALIDATION

# DDIM had lower separability and distributional distance

Both generators matched expression. DDIM was harder to distinguish from real profiles and had higher F1.

## Architecture from Lacan et al.



OSDR adaptation adds FLT/GC, accession and material context through LoRA.

### ARCHS4 tissue probe

Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

| Metric           | WGAN-GP | DDIM  | Read     |
|------------------|---------|-------|----------|
| Correlation      | 0.976   | 0.974 | similar  |
| F1               | 0.985   | 0.997 | higher   |
| Adversarial acc. | 0.636   | 0.475 | near 0.5 |
| FD / real P95    | 0.144   | 0.074 | lower    |

### Use DDIM

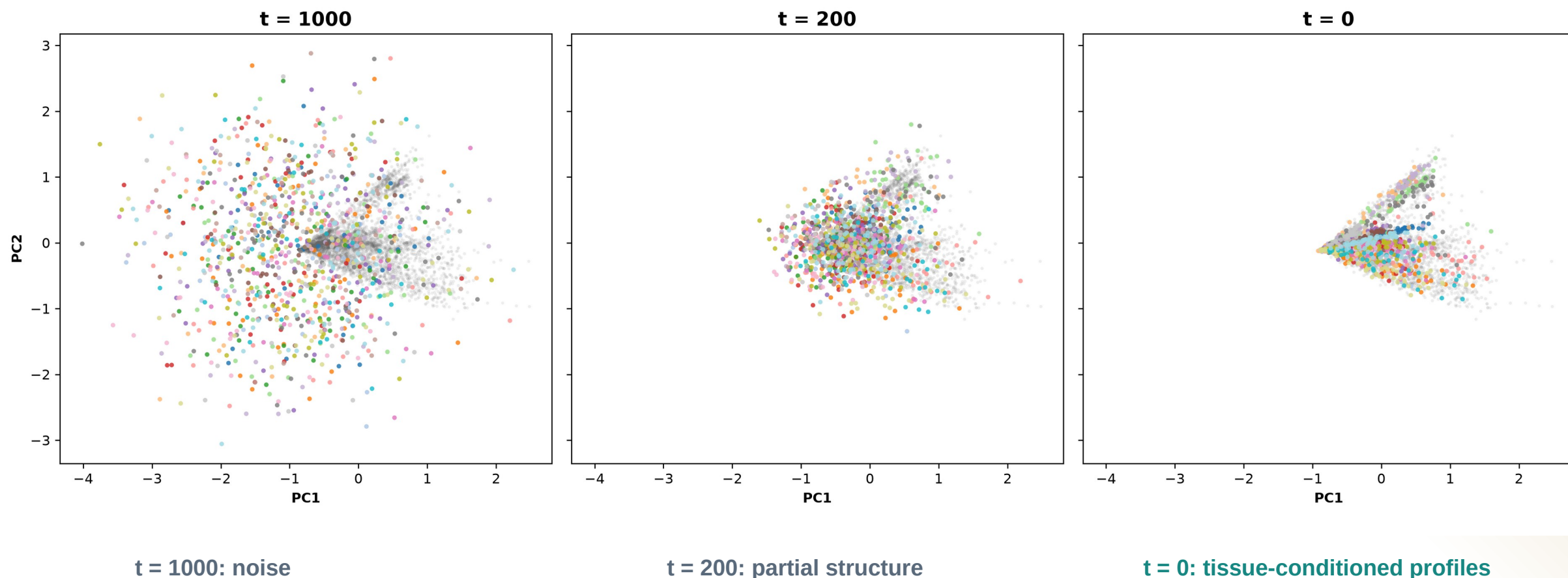
Higher F1 with lower separability and FD.

AA near 0.5 means real and generated profiles are difficult to separate.

## DIFFUSION

# Diffusion learns tissue structure from noise

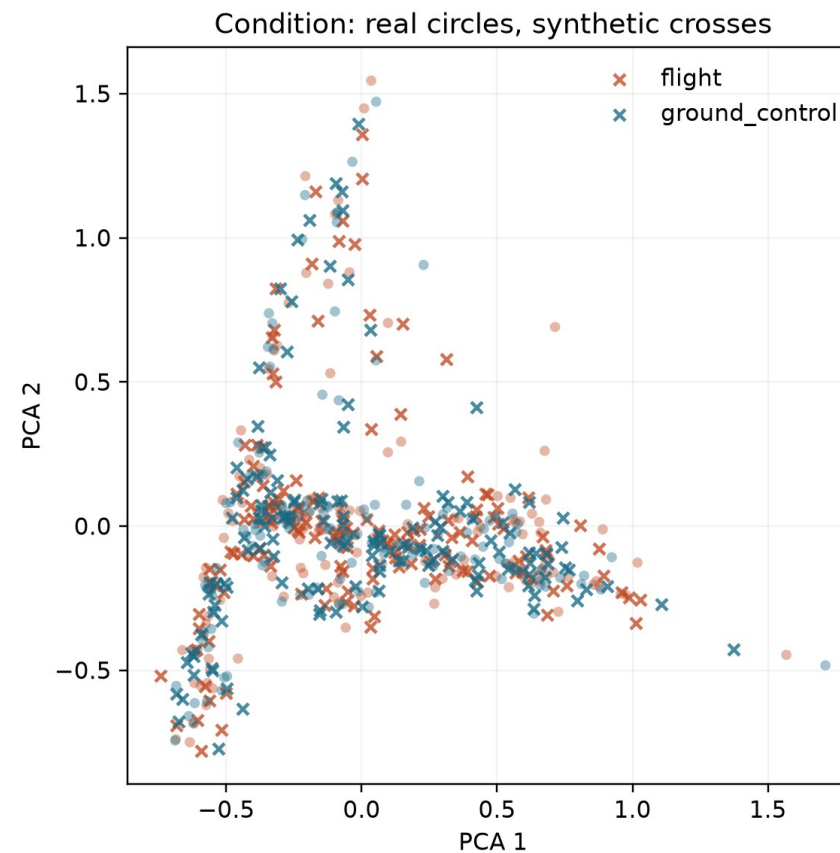
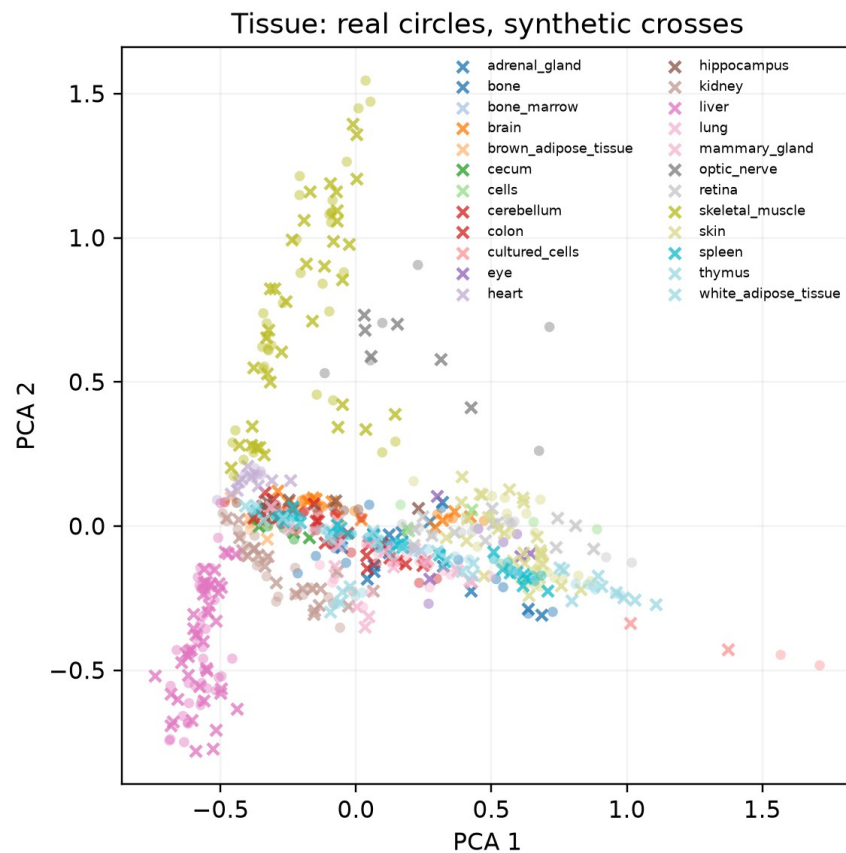
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



## DIFFUSION OUTPUT

# Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.

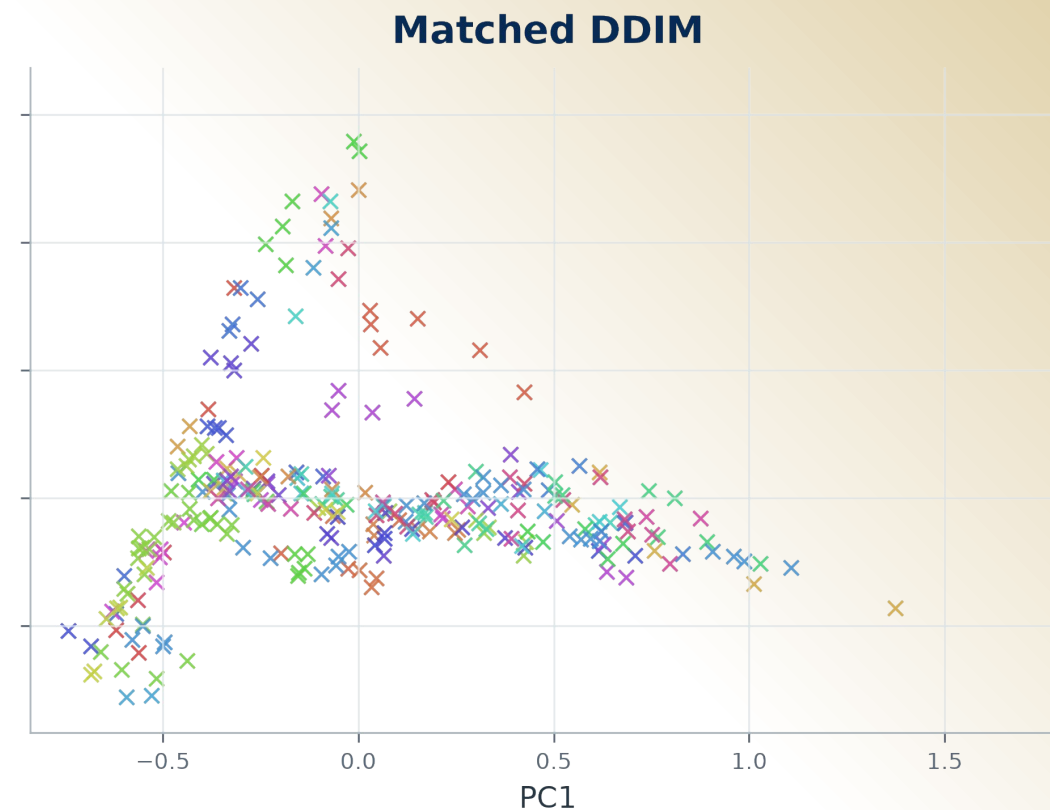
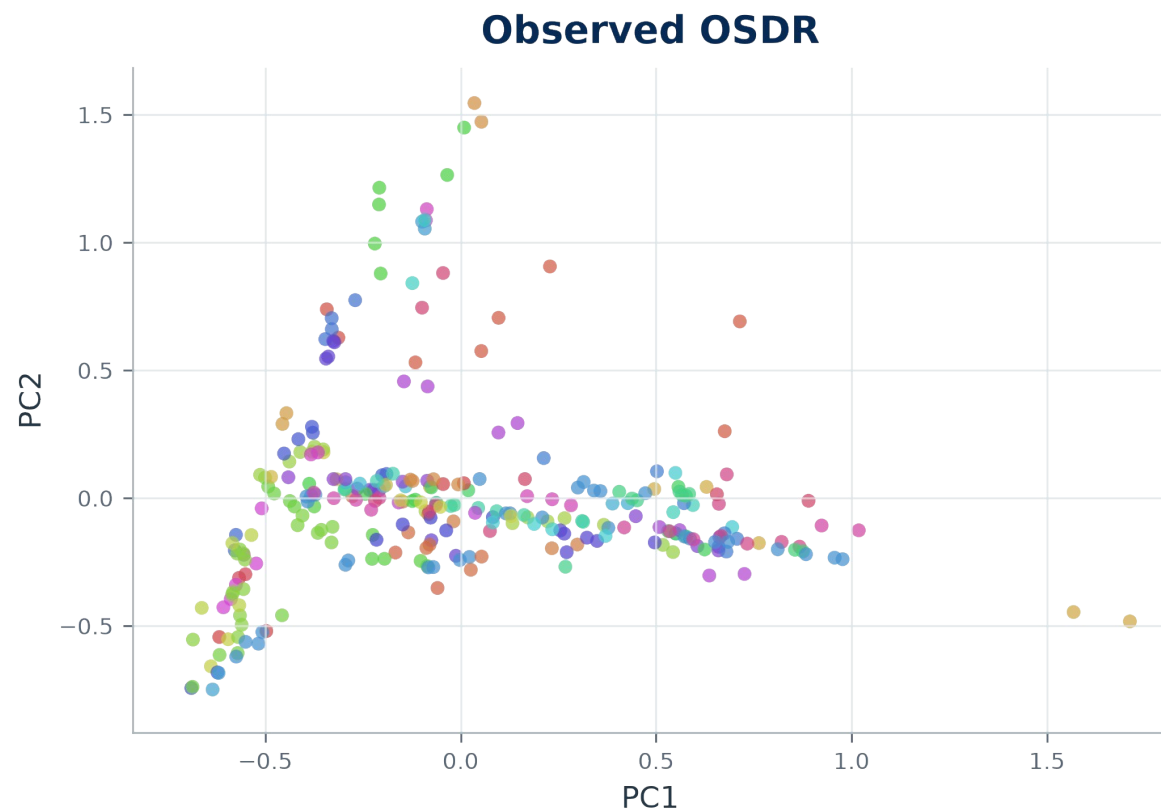


Tissues separate clearly here. Flight and ground-control samples overlap much more, so we test that difference statistically.

## DIFFUSION OUTPUT

# DDIM preserves study-specific structure

The same locked PCA coordinates are recolored by OSDR accession, with one shared color per study.



**Each color is one of 74 OSDR studies. Similar patterns across panels show that the generated profiles retain accession context.**

## ANALYSIS

# Five arms separate gene ranking from classifier fitting

Guided arms use synthetic profiles to help choose genes; they do not treat generated profiles as new animals.

| INPUTS               | <b>R</b> real OSDR    | <b>S</b> matched DDIM | GUIDED                | R + S help rank genes | 5%                              | S carries 5% of total fit weight |
|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|---------------------------------|----------------------------------|
| ANALYSIS ARM         | GENE RANKING          |                       | CLASSIFIER FIT        |                       | WHAT IT TESTS                   |                                  |
| Real only            | <b>R</b> →            | Rank from real        | <b>R</b> →            | Fit real              | Baseline                        |                                  |
| Generated only       | <b>S</b> →            | Rank from generated   | <b>S</b> →            | Fit generated         | Synthetic-only stress test      |                                  |
| Real + generated     | <b>R</b> + <b>S</b> → | Consensus rank        | <b>R</b> + <b>S</b> → | Equal total weight    | Direct augmentation             |                                  |
| Guided: real fit     | <b>R</b> + <b>S</b> → | Consensus rank        | <b>R</b> →            | Fit real only         | Feature guidance only           |                                  |
| Guided: 5% synthetic | <b>R</b> + <b>S</b> → | Consensus rank        | <b>R</b> + <b>S</b> → | S = 5% weight         | Guidance + light regularization |                                  |

All five arms → Held-out real profiles → BA | AUROC | AP → Choose an eligible arm within each tissue

Association test FLT vs GC effects and BH FDR use real OSDR profiles only.

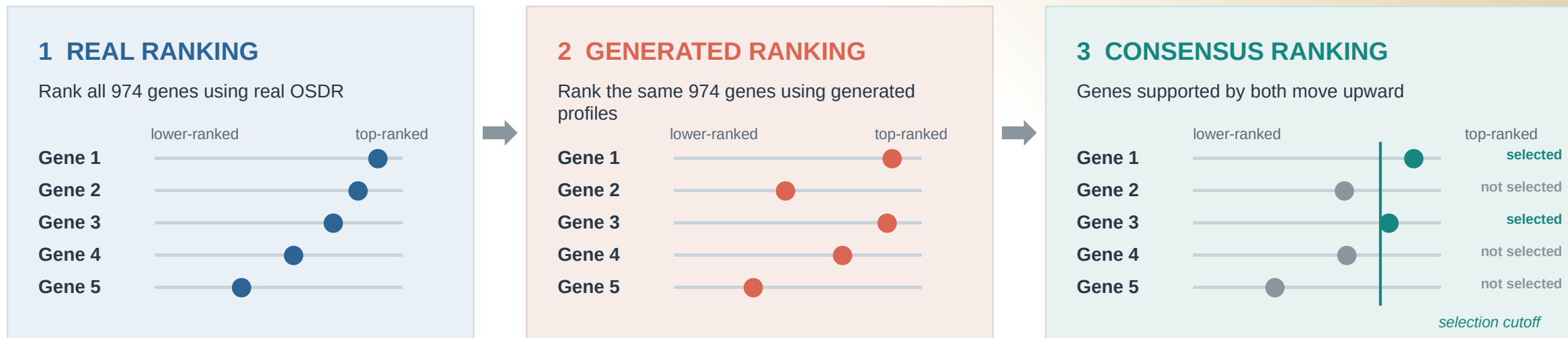
Animal n stays unchanged.



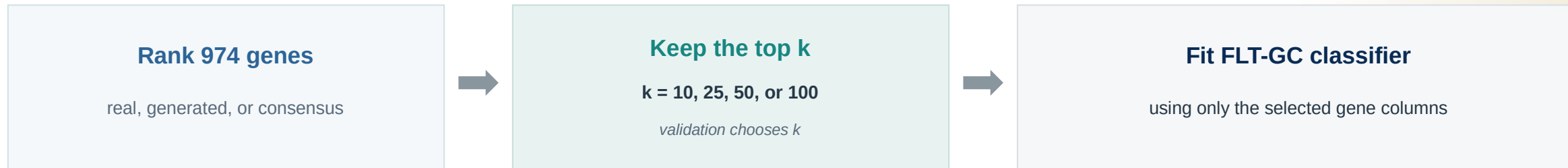
## ANALYSIS

# Consensus ranking chooses the classifier input genes

Real and generated rankings combine, then only the top k genes are used for FLT-GC classification.



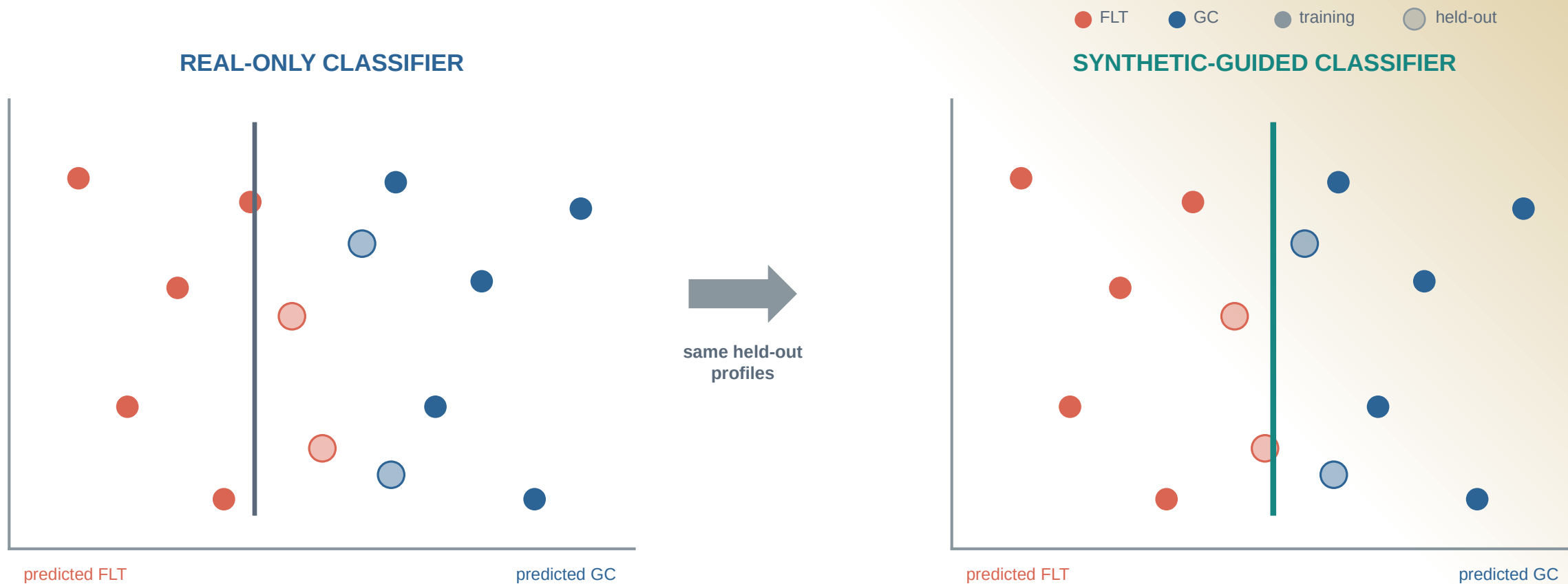
## Top-ranked genes become classifier inputs



## ANALYSIS

# Synthetic guidance can shift the FLT/GC boundary

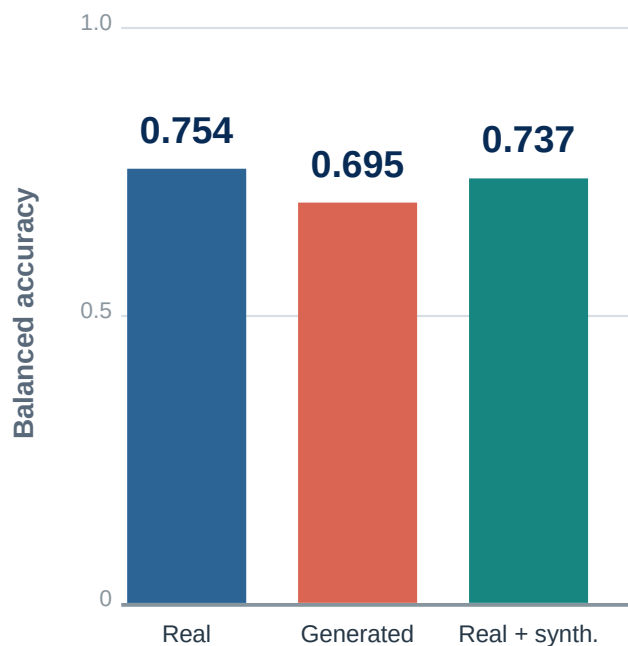
Opaque points train the classifier; transparent points are the same held-out real profiles in both panels.



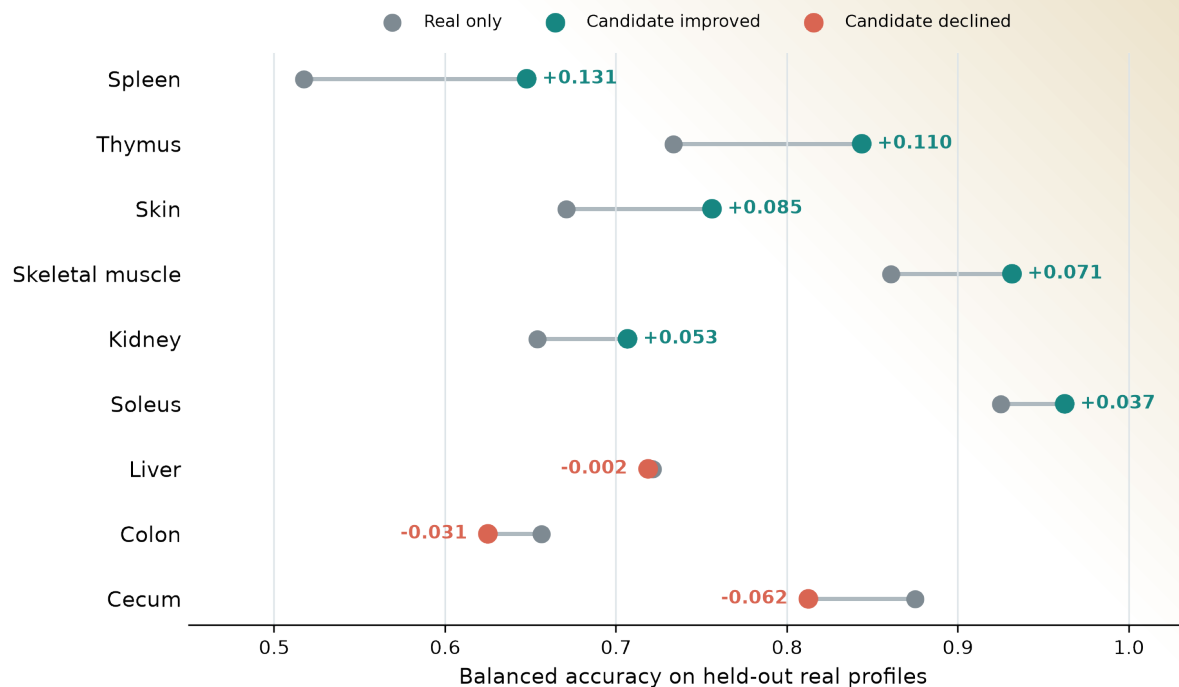
# Synthetic use helped some tissues and hurt others

The pooled classifier declined, and even the best tissue-specific synthetic candidate was not always better than real-only training.

## Pooled across tissues



## Best synthetic-informed candidate by tissue



Only candidates that were non-worse on BA, AUROC, and AP were retained.

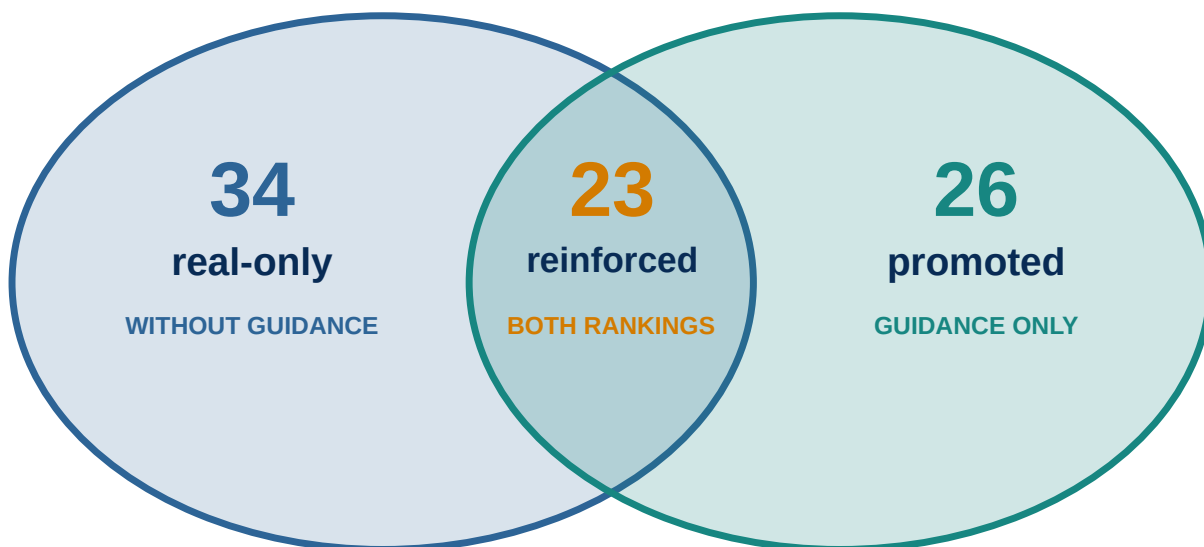
## FEATURE ANALYSIS

# Synthetic guidance changed ranking, not statistical evidence

Blue marks real-only selection; teal marks synthetic-guided selection. Every displayed association has real OSDR support.

## Stable selection after real-data support

- Blue: stable in real-only ranking    ● Teal: stable in synthetic-guided ranking



## Real OSDR evidence gate

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

23 reinforced + 26 promoted  
**49 synthetic-informed associations**

The 34 real-only associations remain real-data findings; synthetic guidance did not reinforce them.

## LITERATURE INTERPRETATION

# Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.

**01 Fix the candidate**

Gene, tissue, and FLT direction

**02 Search**

Spaceflight and mechanism literature

**03 Compare**

Observed result with published evidence

**04 Record**

Category, rationale, and source IDs

## Selection status

**26****Promoted**

Stable only with synthetic guidance

**23****Reinforced**

Stable with and without guidance

## Literature interpretation

| Class                  | Count     | Meaning                                 |
|------------------------|-----------|-----------------------------------------|
| ■ <b>Aligning</b>      | <b>22</b> | Direct or same-tissue process agreement |
| ■ <b>Complementary</b> | <b>19</b> | Related process or mechanism            |
| ■ <b>Ambiguous</b>     | <b>4</b>  | Mixed or context-dependent evidence     |
| ■ <b>Unmatched</b>     | <b>4</b>  | No sufficiently specific match found    |

**A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.**



## ANALYSIS COVERAGE

# The screen covered all 27 completed tissue analyses

The biological discussion narrows only after reporting synthetic-informed, real-only, and null outcomes.

## 10 Synthetic-informed

Promoted or reinforced BH-FDR gene

### ANALYSIS UNITS

- Adrenal Gland
- Eye
- Kidney
- Skeletal muscle (pooled)
- Skin
- Spleen
- Thymus
- Gastrocnemius
- Soleus
- Tibialis Anterior

## 5 Real BH-FDR only

Real association, no synthetic-informed selection

### ANALYSIS UNITS

- Heart
- Liver
- Retina
- EDL
- Quadriceps

## 12 No BH-FDR gene

No BH-FDR gene in the 974-gene panel

### ANALYSIS UNITS

- Bone
- Bone Marrow
- Brain
- Brown Adipose Tissue
- Cecum
- Cerebellum
- Colon
- Hippocampus
- Lung
- Mammary Gland
- Optic Nerve
- White Adipose Tissue

GENE INVENTORY

# Ten tissue analyses contained synthetic-informed genes

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

Rows: FLT direction + status

aligning

complementary

ambiguous

unmatched

Gene color = literature | Non-empty rows shown

|                                                         |                                                                                                                                |  |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--|
| <div>Thymus</div> <div>16 genes</div>                   | <div>FLT higher</div> <div>Promoted</div> <div>Hsd17b11, Etv1</div>                                                            |  |
|                                                         | <div>FLT higher</div> <div>Reinforced</div> <div>Snx7</div>                                                                    |  |
|                                                         | <div>FLT lower</div> <div>Promoted</div> <div>Nusap1, Stmn1, Birc5, Cdk1, Top2a, Ccnb2, Aurka, Ccne2, Kif20a, Pcna, Ccnf</div> |  |
|                                                         | <div>FLT lower</div> <div>Reinforced</div> <div>Ube2c, Gmnn</div>                                                              |  |
| <div>Spleen</div> <div>4 genes</div>                    | <div>FLT higher</div> <div>Promoted</div> <div>Rai14, Myl9, Ptprk</div>                                                        |  |
|                                                         | <div>FLT higher</div> <div>Reinforced</div> <div>Loxl1</div>                                                                   |  |
| <div>Kidney</div> <div>2 genes</div>                    | <div>FLT higher</div> <div>Promoted</div> <div>Inpp4b</div>                                                                    |  |
|                                                         | <div>FLT higher</div> <div>Reinforced</div> <div>Slc37a4</div>                                                                 |  |
| <div>Gastrocnemius</div> <div>2 genes</div>             | <div>FLT higher</div> <div>Promoted</div> <div>Nfkbia</div>                                                                    |  |
|                                                         | <div>FLT lower</div> <div>Promoted</div> <div>Fhl2</div>                                                                       |  |
| <div>Eye</div> <div>1 gene</div>                        | <div>FLT lower</div> <div>Reinforced</div> <div>Klhl21</div>                                                                   |  |
| <div>Skeletal muscle (pooled)</div> <div>12 genes</div> | <div>FLT higher</div> <div>Reinforced</div> <div>Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1</div>                         |  |
|                                                         | <div>FLT lower</div> <div>Promoted</div> <div>Klhl21, Mapkapk5, Reep5, Itgb5</div>                                             |  |
|                                                         | <div>FLT lower</div> <div>Reinforced</div> <div>Bphl</div>                                                                     |  |
| <div>Soleus</div> <div>5 genes</div>                    | <div>FLT higher</div> <div>Reinforced</div> <div>Tpm1</div>                                                                    |  |
|                                                         | <div>FLT lower</div> <div>Reinforced</div> <div>Bdh1, Ech1, Bnip3, Decr1</div>                                                 |  |
| <div>Tibialis anterior</div> <div>4 genes</div>         | <div>FLT higher</div> <div>Promoted</div> <div>Cebpd</div>                                                                     |  |
|                                                         | <div>FLT higher</div> <div>Reinforced</div> <div>Cdkn1a, St3gal5, Bnip3</div>                                                  |  |
| <div>Adrenal gland</div> <div>2 genes</div>             | <div>FLT lower</div> <div>Promoted</div> <div>Psmb8</div>                                                                      |  |
|                                                         | <div>FLT lower</div> <div>Reinforced</div> <div>Tspan4</div>                                                                   |  |
| <div>Skin</div> <div>1 gene</div>                       | <div>FLT higher</div> <div>Promoted</div> <div>Plscr1</div>                                                                    |  |

Separate rows report FLT direction and selection status; gene color independently reports literature interpretation for all 49 associations.

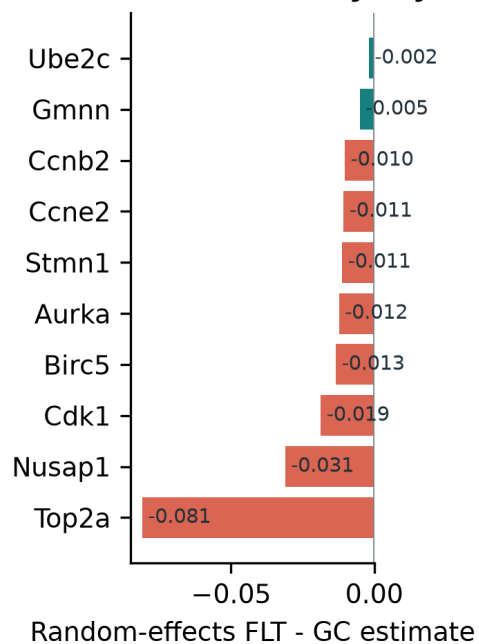
## THYMUS

# Thymus points to lower proliferative renewal

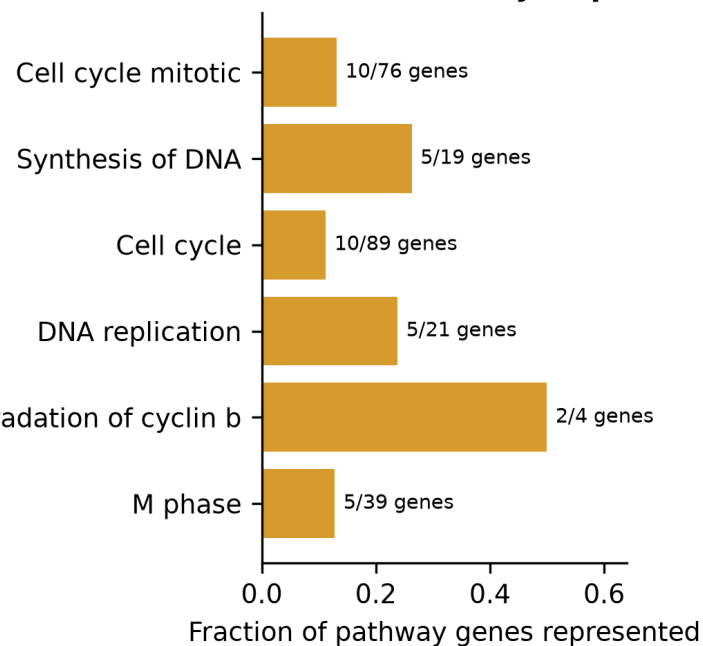
The cell-cycle program persisted without material conditioning; two FLT-higher genes extend the hypothesis.

## Synthetic-informed thymus genes converge on a flight-lower mitotic program

**A Cross-study thymus effects**



**B Shared cell-cycle processes**



16

synthetic-informed  
BH-FDR associations

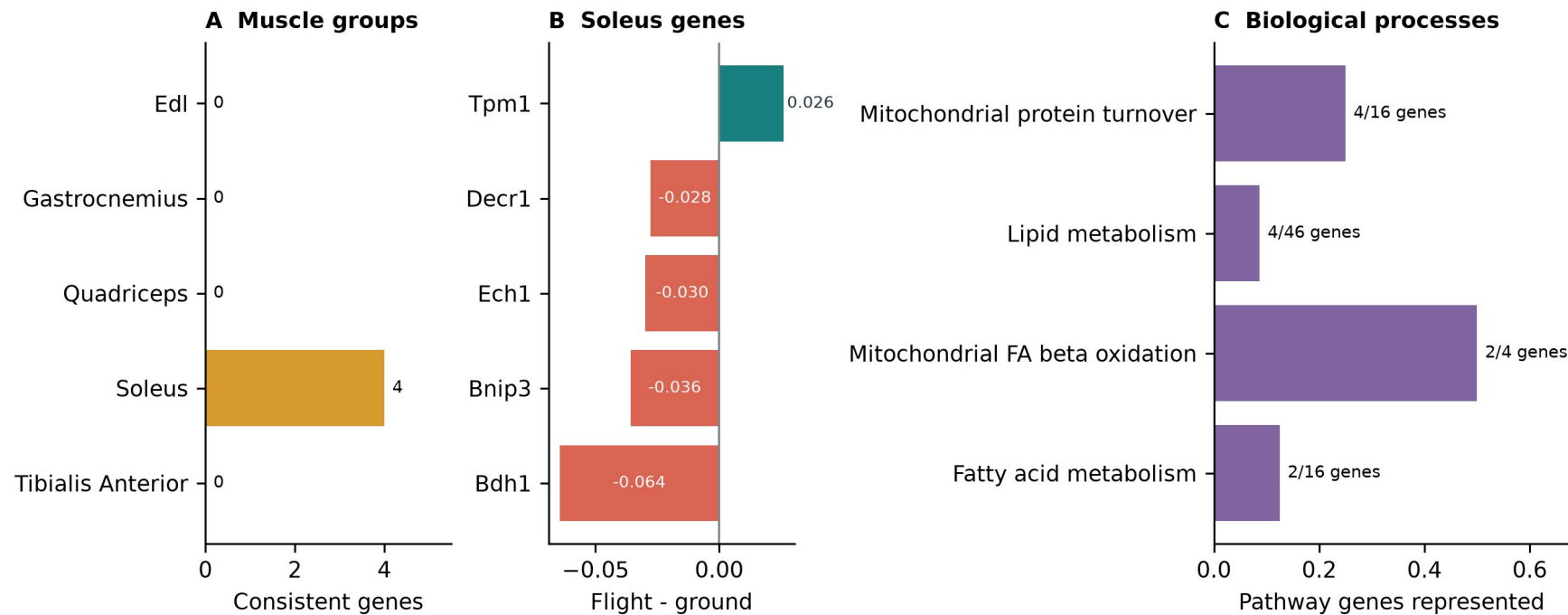
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

# Soleus reinforces a mitochondrial and lipid program

The real-data program is coherent, but its synthetic reinforcement required explicit material conditioning.

## Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



0.925 to 0.963

balanced accuracy

**5 reinforced | 0 promoted**

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

## ADDITIONAL FINDINGS I

# Additional tissues produced distinct hypotheses

Promoted and reinforced selections are shown separately for each tissue.

| Analysis unit | Selection  | Genes and FLT direction                                              | Interpretation                                                                                                           |
|---------------|------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Pooled muscle | Promoted   | Lower: Klhl21, Mapkapk5, Reep5, Itgb5                                | Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups. |
|               | Reinforced | Higher: Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1; lower: Bphl |                                                                                                                          |
| Kidney        | Promoted   | Inpp4b higher                                                        | Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.            |
|               | Reinforced | Slc37a4 higher                                                       |                                                                                                                          |
| Spleen        | Promoted   | Rai14, Myl9, Ptprk higher                                            | Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.         |
|               | Reinforced | Loxl1 higher                                                         |                                                                                                                          |
| Skin          | Promoted   | Plscr1 higher                                                        | Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.               |
|               | Reinforced | None                                                                 |                                                                                                                          |

## ADDITIONAL FINDINGS II

# Eye, adrenal and muscle-group results remain tissue-specific

Promoted and reinforced selections are separated within every tissue.

| Analysis unit     | Selection  | Genes and FLT direction       | Interpretation                                                                                                              |
|-------------------|------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Eye               | Promoted   | None                          | Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication. |
|                   | Reinforced | Klhl21 lower                  |                                                                                                                             |
| Adrenal gland     | Promoted   | Psmb8 lower                   | Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.            |
|                   | Reinforced | Tspan4 lower                  |                                                                                                                             |
| Gastrocnemius     | Promoted   | Nfkb1a higher; Fhl2 lower     | NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.         |
|                   | Reinforced | None                          |                                                                                                                             |
| Tibialis anterior | Promoted   | Cebpd higher                  | Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.              |
|                   | Reinforced | Cdkn1a, St3gal5, Bnip3 higher |                                                                                                                             |



## TAKEAWAYS

# Synthetic data worked best as a tissue-specific prior

The generator helped prioritize observed signal. Biological claims still come from OSDR profiles.

## 01 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

## 02 Use

Tissue-specific consensus ranking and light synthetic regularization improved held-out prediction in selected tissues.

## 03 Interpret

Promoted and reinforced genes supported tissue-specific hypotheses. Literature review separated prior alignment from complementary findings.

# Thank you

Questions?

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Mentor

**James Casaletto**

Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and  
manuscript

[github.com/jasont314/nasa-mouse](https://github.com/jasont314/nasa-mouse)

*All biological associations were tested in observed OSDR profiles.*